



2807 County Rd 75 NE  
Monticello, MN 55362

July 7, 2022

L-MT-22-025  
10 CFR 50.73

ATTN: Document Control Desk  
U.S. Nuclear Regulatory Commission  
Washington, DC 20555-0001

Monticello Nuclear Generating Plant  
Docket No. 50-263  
Renewed Facility Operating License No. DPR-22

LER 2022-001-00, Loss of Control Room Envelope Operability

Pursuant to 10 CFR 50.73(a)(2)(ii)(B), 10 CFR 50.73(a)(2)(v)(D), and 10 CFR 50.73(a)(2)(vii), Northern States Power Company, a Minnesota corporation, doing business as Xcel Energy (hereafter "NSPM"), hereby submits Monticello Nuclear Generating Plant (MNGP) Licensee Event Report (LER) 2022-001-00.

Summary of Commitments

This letter makes no new commitments and no revisions to existing commitments.

A handwritten signature in black ink, appearing to read 'Christopher Domingos', written over a large, stylized 'C'.

Christopher Domingos  
Site Vice President, Monticello and Prairie Island Nuclear Generating Plant  
Northern States Power Company – Minnesota

Enclosure

cc: Administrator, Region III, USNRC  
Project Manager, Monticello, USNRC  
Resident Inspector, Monticello, USNRC  
Minnesota Department of Commerce



## LICENSEE EVENT REPORT (LER)

(See Page 3 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form

<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch (T-6 A10M), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to [Infocollections.Resource@nrc.gov](mailto:Infocollections.Resource@nrc.gov), and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk all: [oir\\_submission@omb.eop.gov](mailto:oir_submission@omb.eop.gov). The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

|                                                         |                                |                   |
|---------------------------------------------------------|--------------------------------|-------------------|
| 1. Facility Name<br>Monticello Nuclear Generating Plant | 2. Docket Number<br>05000 -263 | 3. Page<br>1 OF 3 |
|---------------------------------------------------------|--------------------------------|-------------------|

4. Title  
Loss of Control Room Envelope Operability

| 5. Event Date |     |      | 6. LER Number |                   |              | 7. Report Date |     |      | 8. Other Facilities Involved |               |
|---------------|-----|------|---------------|-------------------|--------------|----------------|-----|------|------------------------------|---------------|
| Month         | Day | Year | Year          | Sequential Number | Revision No. | Month          | Day | Year | Facility Name                | Docket Number |
| 05            | 13  | 2022 | 2022          | - 001 -           | 00           | 07             | 07  | 2022 | Facility Name                | 05000         |
|               |     |      |               |                   |              |                |     |      | Facility Name                | Docket Number |
|               |     |      |               |                   |              |                |     |      |                              | 05000         |

|                        |                        |
|------------------------|------------------------|
| 9. Operating Mode<br>1 | 10. Power Level<br>100 |
|------------------------|------------------------|

## 11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)

|                                                    |                                                    |                                                        |                                                       |                                                    |
|----------------------------------------------------|----------------------------------------------------|--------------------------------------------------------|-------------------------------------------------------|----------------------------------------------------|
| <input checked="" type="checkbox"/> 10 CFR Part 20 | <input type="checkbox"/> 20.2203(a)(2)(vi)         | <input type="checkbox"/> 50.36(c)(2)                   | <input type="checkbox"/> 50.73(a)(2)(iv)(A)           | <input type="checkbox"/> 50.73(a)(2)(x)            |
| <input type="checkbox"/> 20.2201(b)                | <input type="checkbox"/> 20.2203(a)(3)(i)          | <input type="checkbox"/> 50.46(a)(3)(ii)               | <input type="checkbox"/> 50.73(a)(2)(v)(A)            | <input checked="" type="checkbox"/> 10 CFR Part 73 |
| <input type="checkbox"/> 20.2201(d)                | <input type="checkbox"/> 20.2203(a)(3)(ii)         | <input type="checkbox"/> 50.69(g)                      | <input type="checkbox"/> 50.73(a)(2)(v)(B)            | <input type="checkbox"/> 73.71(a)(4)               |
| <input type="checkbox"/> 20.2203(a)(1)             | <input type="checkbox"/> 20.2203(a)(4)             | <input type="checkbox"/> 50.73(a)(2)(i)(A)             | <input type="checkbox"/> 50.73(a)(2)(v)(C)            | <input type="checkbox"/> 73.71(a)(5)               |
| <input type="checkbox"/> 20.2203(a)(2)(i)          | <input checked="" type="checkbox"/> 10 CFR Part 21 | <input type="checkbox"/> 50.73(a)(2)(i)(B)             | <input checked="" type="checkbox"/> 50.73(a)(2)(v)(D) | <input type="checkbox"/> 73.77(a)(1)(i)            |
| <input type="checkbox"/> 20.2203(a)(2)(ii)         | <input type="checkbox"/> 21.2(c)                   | <input type="checkbox"/> 50.73(a)(2)(i)(C)             | <input checked="" type="checkbox"/> 50.73(a)(2)(vii)  | <input type="checkbox"/> 73.77(a)(2)(i)            |
| <input type="checkbox"/> 20.2203(a)(2)(iii)        | <input checked="" type="checkbox"/> 10 CFR Part 50 | <input type="checkbox"/> 50.73(a)(2)(ii)(A)            | <input type="checkbox"/> 50.73(a)(2)(viii)(A)         | <input type="checkbox"/> 73.77(a)(2)(ii)           |
| <input type="checkbox"/> 20.2203(a)(2)(iv)         | <input type="checkbox"/> 50.36(c)(1)(i)(A)         | <input checked="" type="checkbox"/> 50.73(a)(2)(ii)(B) | <input type="checkbox"/> 50.73(a)(2)(viii)(B)         |                                                    |
| <input type="checkbox"/> 20.2203(a)(2)(v)          | <input type="checkbox"/> 50.36(c)(1)(ii)(A)        | <input type="checkbox"/> 50.73(a)(2)(iii)              | <input type="checkbox"/> 50.73(a)(2)(ix)(A)           |                                                    |

☐ OTHER (Specify here, in abstract, or NRC 366A).

## 12. Licensee Contact for this LER

|                                                                       |                                                  |
|-----------------------------------------------------------------------|--------------------------------------------------|
| Licensee Contact<br>Nathan Fedora, Senior Nuclear Regulatory Engineer | Phone Number (Include area code)<br>612-342-8971 |
|-----------------------------------------------------------------------|--------------------------------------------------|

## 13. Complete One Line for each Component Failure Described in this Report

| Cause | System | Component | Manufacturer | Reportable to IRIS | Cause | System | Component | Manufacturer | Reportable to IRIS |
|-------|--------|-----------|--------------|--------------------|-------|--------|-----------|--------------|--------------------|
| B     | SH     | PCV       | C665         | Y                  | X     | SH     | PCV       | C665         | Y                  |

## 14. Supplemental Report Expected

|                                        |                                                                              |                              |       |     |      |
|----------------------------------------|------------------------------------------------------------------------------|------------------------------|-------|-----|------|
| <input checked="" type="checkbox"/> No | <input type="checkbox"/> Yes (If yes, complete 15. Expected Submission Date) | 15. Expected Submission Date | Month | Day | Year |
|                                        |                                                                              |                              |       |     |      |

## 16. Abstract (Limit to 1560 spaces, i.e., approximately 15 single-spaced typewritten lines)

On May 13, 2022, the Monticello Nuclear Generating Plant was operating in Mode 1 at 100% power. Following an inspection in the Steam Jet Air Ejector room, it was identified that the steam leakage from two valves exceeded the assumptions in the Alternate Source Term dose analysis calculation. As a result, the Control Room Envelope (CRE) was declared inoperable and LCO 3.7.4, "Control Room Emergency Filtration (CREF) System," Condition B was entered. The leaks were mitigated such that the CRE was declared operable on May 20, 2022.

This report is being submitted as an "event or condition that resulted in the nuclear power plant being in an unanalyzed condition that significantly degraded plant safety" per 10 CFR 50.73(a)(2)(ii)(B); as an "event or condition that could have prevented the fulfillment of the safety function of structures or systems that are needed to mitigate the consequences of an accident" per 10 CFR 50.73(a)(2)(v)(D); and as "any event where a single cause or condition caused at least one independent train or channel to become inoperable in multiple systems or two independent trains or channels to become inoperable in a single system designed to mitigate the consequences of an accident" per 10 CFR 50.73(a)(2)(vii).

**LICENSEE EVENT REPORT (LER)  
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form  
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

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| 1. FACILITY NAME                    | 2. DOCKET NUMBER | 3. LER NUMBER |                   |         |
|-------------------------------------|------------------|---------------|-------------------|---------|
|                                     |                  | YEAR          | SEQUENTIAL NUMBER | REV NO. |
| Monticello Nuclear Generating Plant | 05000- -263      | 2022          | - 001             | - 00    |

**NARRATIVE****EVENT DESCRIPTION**

On May 13, 2022, the Monticello Nuclear Generating Plant was operating in Mode 1 at 100% power. Following an inspection in the Steam Jet Air Ejector (SJAE) room, it was identified that the steam leakage from two valves exceeded the assumptions in the Alternate Source Term (AST) dose analysis calculation. As a result, the Control Room Envelope (CRE) was declared inoperable and LCO 3.7.4, "Control Room Emergency Filtration (CREF) System," Condition B was entered. The leaks were mitigated such that the CRE was declared operable on May 20, 2022.

**EVENT ANALYSIS**

Steam leakage from the 12 SJAE Pressure Control outlet valve [PCV], MS-23-2, was originally identified on May 5, 2022. As part of an investigation into the leakage, Operators opened the 12 SJAE Pressure Control bypass valve [PCV], MS-24-2, to allow maintenance to remove insulation from MS-23-2. Upon entry into the SJAE room to remove the insulation from MS-23-2 it was discovered that the valve packing from MS-24-2 had failed and was also leaking steam. After removal of the insulation from MS-23-2, it was determined that the leakage from MS-23-2 and MS-24-2 both exceeded the leakage rate assumptions of the AST dose analysis calculation and evaluation. As a result, the Control Room CRE was declared inoperable.

The CRE was inoperable from May 5, 2022 until May 20, 2022 when the leakage sources were mitigated.

This condition was determined to be reportable in accordance with 10 CFR 50.73(a)(2)(ii)(B) as an "event or condition that resulted in the nuclear power plant being in an unanalyzed condition that significantly degraded plant safety," 10 CFR 50.73(a)(2)(v)(D), as an "event or condition that could have prevented fulfillment of a safety function to mitigate the consequences of an accident," and 10 CFR 50.73(a)(2)(vii) as "any event where a single cause or condition caused at least one independent train or channel to become inoperable in multiple systems or two independent trains or channels to become inoperable in a single system designed to mitigate the consequences of an accident."

This event will be reported as a Safety System Functional Failure in accordance with Nuclear Energy Institute 99-02, "Regulatory Assessment Performance Indicator Guideline".

**SAFETY SIGNIFICANCE**

There was no actual safety consequence from this condition. During the time the CRE was inoperable radiation levels in the Turbine Building and on the Wide Range Gas Monitors remained stable. Compensatory measures were put in place, including assigning a dedicated operator to promptly isolate the portion of the SJAE system that was leaking in the event of a design basis event. In addition, the control room operators were directed to don a self-contained breathing apparatus (SCBA) on breathing air if a design basis event were to occur. An engineering analysis determined that donning SCBAs would ensure control room operator doses remained below 5 Rem Total Effective Dose Equivalent (TEDE). Prior to the implementation of the compensatory measures described above on May 13, 2022, it is estimated the post-accident dose limit for the control room operators could have been exceeded. During the time the steam leakage in the SJAE Room exceeded the assumptions of the AST analysis both CREF subsystems were otherwise available.

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**NARRATIVE****CAUSE**

The direct cause of the CRE being declared inoperable was that the accumulated leakage had exceeded the leakage rate assumptions of the AST dose analysis calculation. The margin to the assumed AST leakage was eroded by the leaks which were identified previous to MS-23-2 and MS-24-2. The additional steam leakage from MS-23-2 and MS-24-2 exceeded the leakage rate assumptions of the AST dose analysis calculation. The cause of the steam leak from MS-24-2 was improperly consolidated valve packing when the valve was installed. MS-23-2 developed a through body leak, the apparent cause of which cannot be determined without an inspection of the valve body.

**CORRECTIVE ACTIONS**

The leakage from MS-23-2 and MS-24-2 has been contained. Both valves will be replaced during the next refueling outage. Procedural clarifications are being developed to verify valve packing is properly installed and consolidated. In addition, the current maintenance strategies will be reviewed against the consequences associated with AST.

**PREVIOUS SIMILAR EVENTS**

There were no previous similar licensee event reports in the past three years.

**ADDITIONAL INFORMATION**

The Institute of Electrical and Electronics Engineer codes for equipment are denoted by [XX].